

Hebbian learning

Hebbian plasticity in the broadest sense is the idea that the correlations between the spiking activity of pre- and post- synaptic neurons determine the synaptic weight between them. In

Let us assume that the persistence or repetition of a reverberatory activity (or “trace”) tends to induce lasting cellular changes that add to its stability. ...When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A’s efficiency, as one of the cells firing B, is increased. [2]

The synaptic plasticity rule most similar to Hebb’s intuition is called **Spike-Timing Dependent Plasticity** (STDP). In the most common form STDP, if the presynaptic cell causally contributes to firing the postsynaptic cell, the synapse gets stronger. If the presynaptic cell routinely fires late (after postsynaptic cell has already fired), the synapse gets weaker. Experiments have observed synapses between pairs of excitatory neurons (approximately) following this rule.

Hebbian learning is also described as “neurons that fire together wire together”. Hebb never said this. This interpretation ignores causality, but it is an OK summary of the learning rule in simple Hopfield networks we explored, which model area CA3 in the hippocampus as an auto-associative network with Symmetric weights. In the Hopfield lab, we make a positive synapse for correlated “neurons” $(-1,-1 \text{ or } 1,1)$ and a negative synapse for anti-correlated ones $(1,-1 \text{ or } -1,1)$. We don’t model which neurons spike first.

Other neurons have different learning rules. Inhibitory neurons in particular have important roles in regulating neural activity, and since they are trying to make the post-synaptic cell *not* spike, they sometimes have learning rules described as “anti-Hebbian”. There are other variants, but we will not dig into these too deep.

Since these details are normally discussed later in the course, they are not examined to rigorously on the midterm. The main benefit I see is that explaining these details might make it a bit more clear about what we imagine is happening in the CA3 autoencoder or cerebellar Purkinje cell “perceptron” architecture. It might also help contextualize the labs, which might

seem a bit silly and abstract without the biological context above.

References

- [1] Hopfield, JJ. (1982) Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences of the USA*, 79:2554–2558.
- [2] Hebb DO. (1949) *The Organization of Behavior*. New York: Wiley & Sons.
- [3] Amit D. (1992) *Modeling Brain Function: The World of Attractor Neural Networks*. Cambridge University Press, Cambridge England.